Practice Quiz: Creative Action

Part A: Multiple Choice

1. In Active Inference, creative action serves two simultaneous functions: A) Revenue generation and cost reduction B) Pragmatic inference (changing the world) and epistemic inference (generating observations) C) Impressing stakeholders and meeting deadlines D) Testing materials and selecting vendors

2. Fidelity matching in prototyping means: A) Making every prototype as realistic as possible B) Matching prototype fidelity to the specific question being tested C) Using the same materials as the final product D) Building prototypes that match competitor products

3. James Dyson's 5,127 prototypes are best understood as: A) 5,127 failures followed by one success B) 5,127 cycles of the perception-action loop, each generating informative prediction errors C) Evidence that Dyson lacked a coherent design methodology D) An unusually inefficient development process

4. Bricolage differs from conventional engineering in that: A) It is less creative and produces inferior results B) Available materials shape the design rather than a predetermined design specifying materials C) It requires more expensive equipment D) It is only used when proper materials are unavailable

5. The design sprint's five-day time constraint is valuable because: A) It reduces labor costs B) It forces action over analysis, compressing the perception-action-cognition cycle C) Shorter projects always produce better results D) It allows team members to return to other projects sooner

6. Jeff Hawkins developed the Palm Pilot form factor using: A) Advanced computer-aided design software B) A carved wooden block carried in his pocket for months to simulate usage C) Focus groups and market research surveys D) A fully functional electronic prototype from day one

7. The Apollo 13 CO_2 scrubber repair exemplifies: A) Standard engineering practice with proper materials B) The failure of planned designs under real-world conditions C) Bricolage under extreme constraint — invention using only available materials D) The superiority of prefabricated solutions over improvisation

Part B: Short Answer / Analysis

1. A startup team has spent three months developing detailed CAD models and engineering specifications for a new ergonomic computer mouse, but has not yet built a single physical prototype. Using the Active Inference framework, explain what they are missing and propose a prototyping strategy. Specify three prototypes at different fidelity levels, each testing a different critical assumption, with a timeline of no more than two weeks.

2. The module argues that "making is cognition, not just execution." Provide two specific examples (one from the module's case studies and one from your own experience or imagination) where hands-on construction revealed an insight that no amount of mental simulation could have produced. For each example, explain what type of information the physical action generated that mental simulation lacked.

3. You are developing a new type of portable water filtration device for use in regions without clean water infrastructure. Design a bricolage prototyping session: list the everyday materials you would use (at least eight items), describe how you would construct a functional prototype from these materials, and explain what specific questions this prototype would answer about your invention's viability. Include at least two different prototype iterations and what you expect to learn from each.